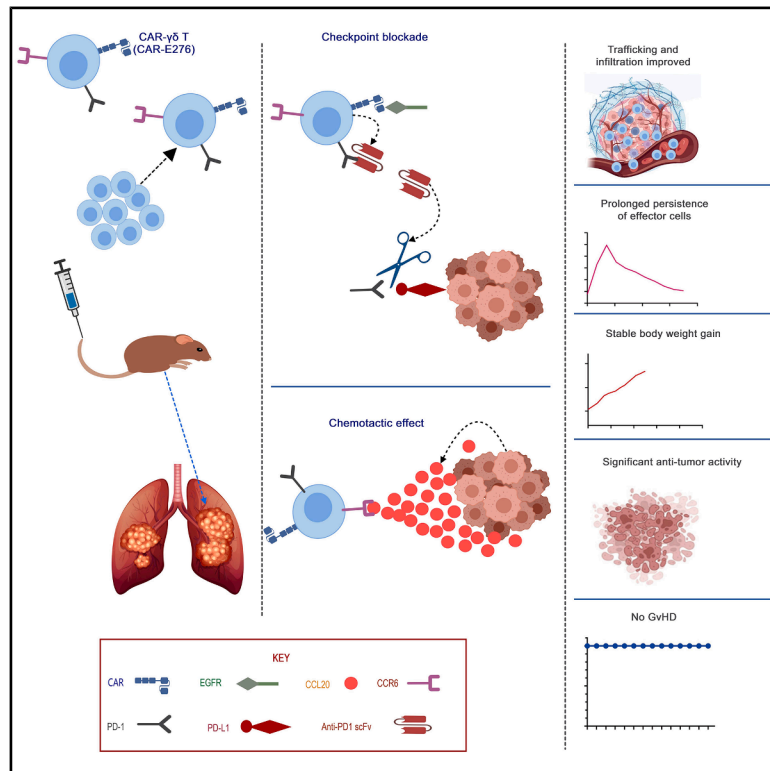


Validation of CCL20-driven CAR- $\gamma\delta$ T secreting PD-1 blockade with enhanced trafficking into solid tumor

Graphical abstract



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In brief

Cellular therapy; Immune response;
Cancer

Highlights

- CAR-E276 can be efficiently delivered to solid tumors via systemic administration
- CAR-E276 exhibits prolonged persistence and improves survival of treated animals
- CAR-E276 demonstrates robust anti-tumor activity *in vivo* without inducing GvHD
- CAR-E276 shows promise as a universal, allogeneic, off-the-shelf cell therapy product



Article

Validation of CCL20-driven CAR- $\gamma\delta$ T secreting PD-1 blockade with enhanced trafficking into solid tumor

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SUMMARY

The primary challenges of CAR-T cells for solid tumor treatment involve the efficient delivery and infiltration of CAR-T cells into the tumor site, as well as overcoming the immunosuppressive tumor microenvironment. Combining chemokine-guided trafficking and delivery with the clearance of immunosuppressive barriers represents a promising strategy. An EGFR-targeted CAR- $\gamma\delta$ T, CAR-E276 that co-expresses CCR6 and secretes PD1 blockade was *in vivo* validated using a non-small cell lung cancer (NSCLC) CDX model. The results from CAR-E276 indicated that integrating chemokines, checkpoint blockades, and $\gamma\delta$ T cell properties into CAR-T cells facilitated their efficient trafficking into tumor tissues, ensured prolonged persistence, and achieved robust tumor-killing effects without inducing GvHD. Notably, CAR-E276 also exhibited potential for off-the-shelf and allogeneic applications. These findings are expected to offer valuable insights and drive the development of CAR-T therapies targeting solid tumors.

INTRODUCTION

Cell and gene therapy (CGT) drugs represent a burgeoning field of drug development following traditional chemical drugs and large molecule drugs. CGT drugs encompass a broad spectrum, including mRNA drugs, oligonucleotide drugs, and CAR-T cell therapies. The efficacy of CGT drugs often depends on delivery systems that provide protection and/or targeting capabilities. For instance, mRNA drugs currently available on the market require lipid nanoparticle (LNP) encapsulation for efficient delivery, while oligonucleotide drugs rely on LNP encapsulation or GalNAc modification to achieve targeted delivery. In contrast, CAR-T therapies do not necessitate an independent delivery system but instead achieve targeted delivery through genetic modification of chimeric antigen receptors (CARs). The initial design of CAR-T therapies primarily relied on single-chain variable fragments (scFvs) targeting tumor-associated antigens (TAAs) or tumor-specific antigens (TSAs) to facilitate targeted delivery and interaction with tumors. This approach is relatively straightforward and feasible for hematological malignancies but encounters significant obstacles when extended to the treatment of solid tumors. To effectively treat solid tumors, CAR-T therapies should incorporate more sophisticated strategies that address both targeted delivery and tumor infiltration, rather than relying solely on tumor antigen recognition.

To date, a total of 11 CAR-T products have received global approval, with 6 approved by the Food and Drug Administration (FDA) and 5 authorized in China. These products are primarily indicated for hematological malignancies, including lymphoma, myeloma, and leukemia. However, no CAR-T product has yet been successfully approved for the treatment of solid tumors, highlighting the need for substantial breakthroughs in this area. Furthermore, all approved CAR-T therapies are expensive autologous treatments. To successfully translate CAR-T therapy for solid tumors into clinical application, several significant challenges must be addressed. These challenges are predominantly attributed to the complex nature of the tumor microenvironment (TME). The TME presents significant obstacles, such as physical barriers that hinder CAR-T cell delivery and trafficking, as well as an immunosuppressive environment that suppresses CAR-T cell activity.¹ Cancer cells limit antitumor responses and affect effector cells' persistence by expressing immune checkpoints. One well-known immune checkpoint inhibition pathway is PD1/PD-L1, where the complex formed by PD1 and PD-L1 inhibits T cell cytotoxicity, thereby creating an immunosuppressive environment.² The tumor microenvironment is replete with inhibitory PD-L1 signaling, because PD-L1 can be expressed on both tumor cells and host cells, including fibroblasts,³ dendritic cells, macrophages, and B and T cells. PD-L1 expression may reflect an adaptive cancer response to immune infiltration or occur independently of immune cell activation.⁴



The ineffective delivery of CAR-T cells to tumor site and their insufficient infiltration in the solid tumor microenvironment can be attributed to the physical barrier, which represents another main challenge in this context. Enhancing CAR-T cell trafficking into the tumor microenvironment is essential for augmenting anti-tumor immune responses. Chemokines play a pivotal role in recruiting and signaling immune cells, potentially promoting the migration of effector T cells and natural killer (NK) cells into tumors.⁵ Several Studies have demonstrated that improved therapeutic responses to cancer immunotherapy and chemotherapy are associated with increased levels of chemokines and increased numbers of effector T cells in the tumor microenvironment.^{6,7} Epigenetic silencing of genes encoding chemokines, such as CXCL9 and CXCL10, results in reduced T cell infiltration into tumors.^{8,9} The presence of CCL20 in human tumor microenvironments suggests that this chemokine profile may facilitate T cell trafficking into tumors.¹⁰ Unlike other chemokines, current studies have shown that CCL20 binds specifically to C-C chemokine receptor type 6 (CCR6) and the expression of CCL20 in lung cancer tissues is significantly higher than that in normal lung tissues, which are helpful for us to make use of them.¹¹

There are various types of immune cells that can be utilized for constructing CAR-T cells, each with distinct advantages. The most used cells for CAR-T construction are $\alpha\beta$ T cells; however, $\alpha\beta$ T cells are primarily suited for autologous therapy and may easily induce graft-versus-host disease (GvHD), which poses challenges for constructing allogeneic CAR-T cells. Compared with $\alpha\beta$ T cells, although $\gamma\delta$ T cells account for only 1%–5% of total T cells,¹² they do not recognize MHC molecules and lack homologous reactivity.^{13–15} These characteristics render $\gamma\delta$ T cells a promising candidate for engineering universal allogeneic CAR-T cells, potentially reducing the economic burden on patients. Additionally, according to Harrer et al.'s study, CAR-transfected $\gamma\delta$ T cells produce lower levels of cytokines compared to CAR-transfected $\alpha\beta$ T cells, thereby reducing the risk of cytokine release syndrome (CRS).¹⁶ Another significant advantage is that $\gamma\delta$ T cells not only lack MHC restriction but also exhibit natural tissue tropism. This enables them to migrate more effectively to tissues and infiltrate solid tumors compared to traditional $\alpha\beta$ T cells, even perform their functions well under hypoxic conditions.¹⁷

To combat the complex microenvironment of solid tumors, beyond employing CAR to recognize TSAs or TAAs, more comprehensive and integrated strategies are warranted. These encompass harnessing the chemotactic effects of chemokines to overcome the migration and infiltration obstacles encountered by effector cells when accessing tumor tissues, as well as simultaneously using immune checkpoint blockades to alleviate immunosuppressive microenvironmental barriers. Such strategies create a synergistic effect where immune checkpoint blockades remove roadblocks while chemokines enhance recruitment. Building on the favorable tissue tropism of $\gamma\delta$ T cells as seed cells, these strategies may hold significant potential for synergistically promoting CAR-T cell delivery, infiltration, and long-term persistence within tumor tissues, ultimately enhancing therapeutic efficacy.

In this study, we evaluated CAR-E276, an engineered CAR-T targeting EGFR with co-expression of CCR6 and secretion of anti-PD1 scFv. This CAR-T was developed using $\gamma\delta$ T cells as

seed cells and employed multiple strategies to enhance CAR-T activity as described previously. To validate the effectiveness of this CAR-T construction strategy, we selected non-small cell lung cancer (NSCLC) as its first indication and established a CDX model to comprehensively assess the *in vivo* distribution, persistence, and cytotoxic activity of CAR-E276. Our *in vivo* study data demonstrated that the integration of chemokines, immune checkpoint inhibitors, and $\gamma\delta$ T cells facilitated CAR-T delivery to tumor tissues, promoted longer-term persistence, and achieved significant tumor growth inhibition without inducing GvHD. Furthermore, CAR-E276 exhibits potential for off-the-shelf and allogeneic applications. We anticipate that our findings will inspire promising approaches for constructing allogeneic CAR-T therapies targeting solid tumors.

RESULTS

Packaging and identification of lentivirus containing transgenes CCL20 and CD274

To evaluate the *in vivo* activity of CAR-T, it is essential to construct tumor cell lines for the establishment of CDX animal models. Currently, tumor cell lines are typically established via lentivirus transfection, as lentiviruses can integrate into the cellular genome and enable stable expression of the transgenes. We utilized the shuttle plasmid pHBLV-CMV-MCS-EF1-fLUC-T2A-Puro, which contains the luciferase gene, to transfer gene of CD274 and CCL20 according to the schematic diagram (Figure 1A) and constructed an expression vector. Subsequently, co-transfection with the psPAX2 and pMD2G auxiliary plasmids enabled the packaging of 293T cells to produce a new lentivirus for the subsequent construction of the intended tumor cell line. The infectivity of different viral titers was assessed by culturing 293T cells in medium supplemented with PURO (Figures 1B–1E). The titers of the newly packaged virus HBLV-h-CD274-P2A-CCL20-LUC-PURO carrying the target gene and its control virus HBLV-LUC-PURO were determined to be 2×10^8 TU/mL.

Construction and identification of NSCLC cell line expressing CCL20, CD274, EGFR, and luciferase

Tumor cell lines are indispensable for *in vivo* efficacy studies of CAR-T cells and serve as crucial tools for establishing CDX models. Standardized tumor cell lines with stable expression of intended transgenes facilitate more reliable *in vivo* validation of CAR-T cell efficacy. Given that CAR-T cells recognize the extracellular domain of TAAs or TSAs, and upon activation, they exert cytotoxic effects on target cells through the release of granzymes and perforin, activation of the FAS pathway, and secretion of cytokines.^{18,19} The mechanism of action of CAR-T therapy is significantly different from that of small molecule targeted therapy for NSCLC. This difference gives CAR-T therapy a key advantage, namely its ability to effectively target tumors with wild-type EGFR and mutant EGFR. Therefore, choosing a NSCLC cell line with high expression of EGFR and containing the expected targeting domain of CAR-T is suitable for this research. The NSCLC cell line A549 was established by Gerard et al. in 1972 and originated from male Caucasian lung cancer tissue. This cell line represents the wild-type cells expressing

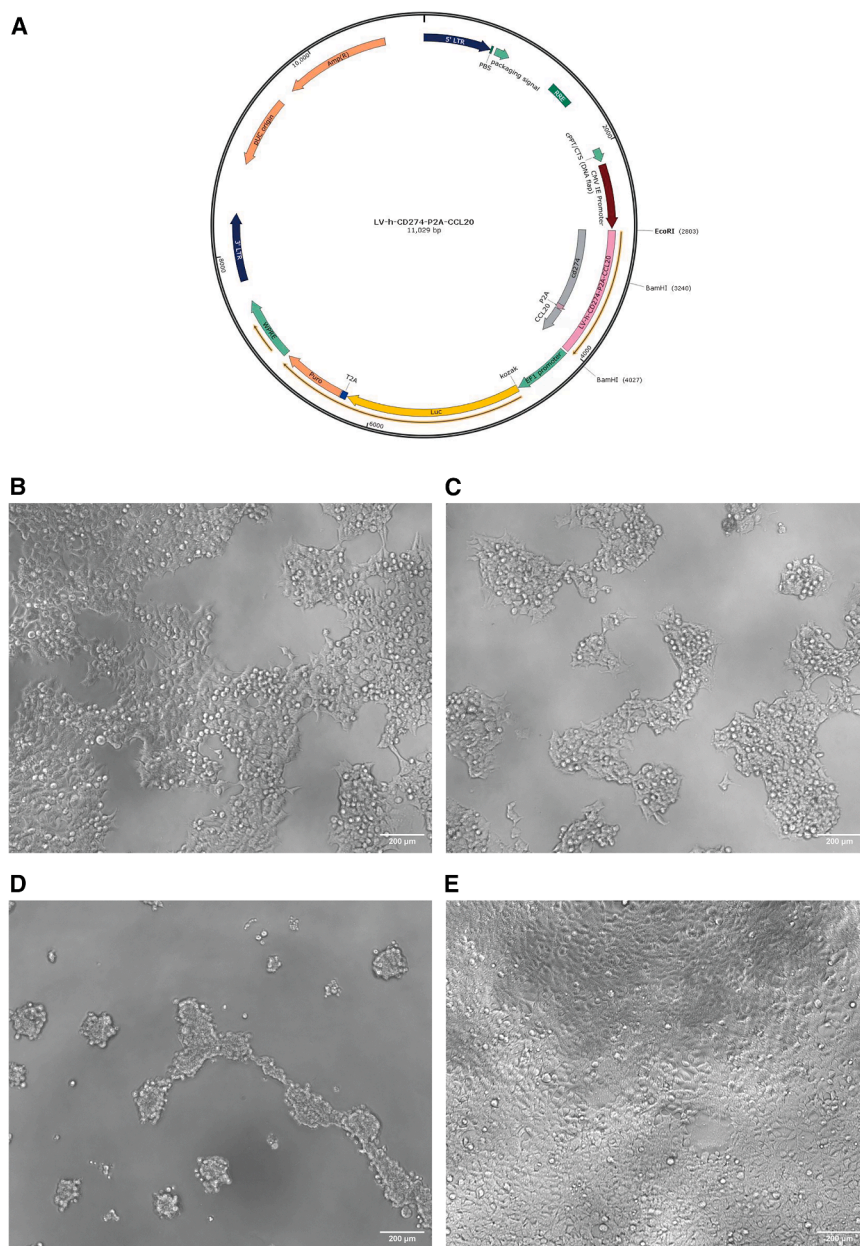


Figure 1. Lentivirus packaging and identification of infectivity capability

(A) Construction map of the expression plasmid. (B) The results of the infection experiment conducted with 0.11 μ L of newly packaged virus on 293T cells cultured in medium supplemented with puromycin. (C) The results of the infection experiment conducted with 0.033 μ L of newly packaged virus on 293T cells cultured in medium supplemented with puromycin. (D) The control experiment results of uninfected 293T cells cultured in medium containing puromycin. (E) The control experiment results of uninfected 293T cells cultured in puromycin-free medium. All the micrographs were taken at a magnification of 10×10 . Scale bars, 200 μ m.

titative RT-qPCR analysis revealed a significant increase in CD274 and CCL20 mRNA expression levels in A549-CD274-CCL20-Luc cells (Figures 2A and 2B). Furthermore, we validated the significant expression of these transgenes at the protein level in the newly constructed NSCLC cell line. Given that CCL20 is a secreted protein (also known as MIP-3 α), we collected the cell culture supernatant after 24 h of incubation and used ELISA to detect its expression level. The newly constructed tumor cell line (A549-CD274-CCL20-Luc) exhibited significantly higher CCL20 expression compared to control A549 cells (Figure 2C). Since CD274 is a membrane protein, we confirmed its expression in A549-CD274-CCL20-Luc cells using flow cytometry with AF488-labeled anti-CD274 antibodies (Figure 2D), comparing it to non-transgenic A549 cells. We also verified that the newly established cell line retains EGFR expression by staining with PE-labeled anti-EGFR antibodies and analyzing via flow cytometry (Figure 2E).

EGFR and has already been utilized in several CAR-T studies.^{20–23} Therefore, we only needed to introduce the CCL20, CD274 (PD-L1), and luciferase genes into this cell line. To establish a stable NSCLC cell line expressing CD274, CCL20, and luciferase, we transfected A549 cells with a newly constructed lentivirus carrying CD274, CCL20, and luciferase genes. Subsequently, we employed puromycin to screen the transfected cells, thereby successfully establishing an A549 cell line that stably expresses CD274, CCL20, and luciferase. This cell line was designated as A549-CD274-CCL20-Luc.

We identified and confirmed the expression of PD-L1 (CD274) and CCL20 genes in the newly established A549-CD274-CCL20-Luc cell line. Compared with control cell line (A549), relative quan-

titative RT-qPCR analysis revealed a significant increase in CD274 and CCL20 mRNA expression levels in A549-CD274-CCL20-Luc cells (Figures 2A and 2B). Furthermore, we validated the significant expression of these transgenes at the protein level in the newly constructed NSCLC cell line. Given that CCL20 is a secreted protein (also known as MIP-3 α), we collected the cell culture supernatant after 24 h of incubation and used ELISA to detect its expression level. The newly constructed tumor cell line (A549-CD274-CCL20-Luc) exhibited significantly higher CCL20 expression compared to control A549 cells (Figure 2C). Since CD274 is a membrane protein, we confirmed its expression in A549-CD274-CCL20-Luc cells using flow cytometry with AF488-labeled anti-CD274 antibodies (Figure 2D), comparing it to non-transgenic A549 cells. We also verified that the newly established cell line retains EGFR expression by staining with PE-labeled anti-EGFR antibodies and analyzing via flow cytometry (Figure 2E).

Luciferase expression in newly tumor cell line was also confirmed (Figure 2F). Mycoplasma contamination represents a critical threat to cell growth and is a prevalent source of contamination in cell culture. If mycoplasma contaminates engineered cells, it can significantly impact the downstream applications of the cell line, especially in the animal modeling based on this tumor cell line. Mycoplasma contamination can hinder tumor formation in NSCLC-CDX mouse models and, even if tumors do form, severely compromise the uniformity of tumor growth, thereby affecting the quality of the final tumor-bearing mouse models. For studies involving the development of products intended for industrial applications, it is imperative to eliminate mycoplasma

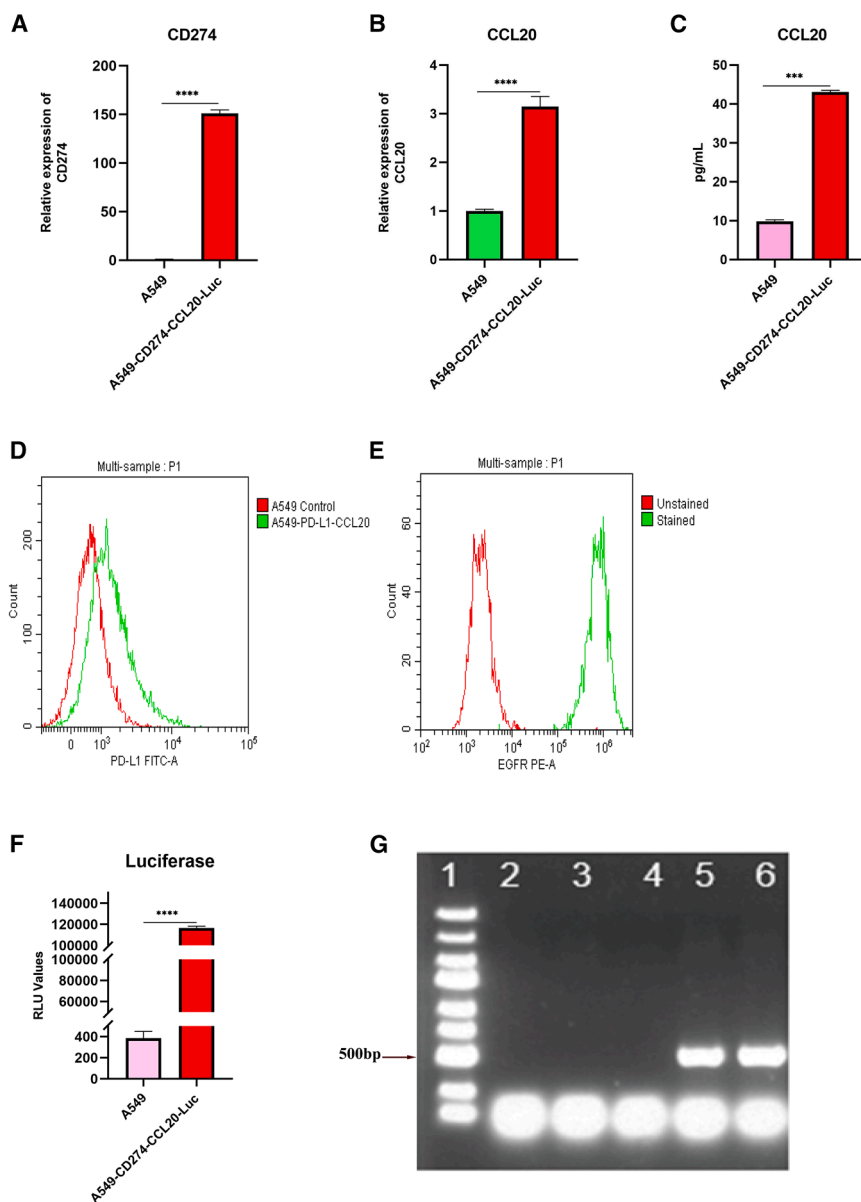


Figure 2. Quality assessment and validation of A549-CD274-CCL20-Luc cells

(A and B) RT-qPCR identification of CD274 and CCL20 expression in the A549-CD274-CCL20-Luc cells. Data are represented as mean $\pm$ SD ($n = 3$, independent experiments).

(C) ELISA analysis of CCL20 expression in the A549-CD274-CCL20-Luc cells. Data are represented as mean $\pm$ SD ($n = 2$, independent experiments).

(D and E) The expression of PD-L1 and EGFR were identified by flow cytometry, respectively.

(F) Luciferase expression was confirmed by luciferase assay. Data are represented as mean $\pm$ SD ($n = 3$, independent experiments).

(G) Mycoplasma contamination was assessed using PCR, and the PCR results were identified by agarose gel electrophoresis. Lane 1 is DNA marker, lane 2 is a non-transfected control cell (A549), lane 3 is a newly constructed A549-CD274-CCL20-Luc cell, lane 4 is a negative control, and lane 5 and lane 6 are positive controls, a band of approximately 500 bp will appear in the sample if contaminated with mycoplasma.

Statistical analyses were performed using the two-tailed unpaired Student's *t* test. *** $p < 0.001$, **** $p < 0.0001$.

contamination from the cell lines used in model establishment to ensure the generation of high-quality mouse models with consistent tumor growth. In this study, we pre-screened the lung cancer cell line A549 engineered to express CCL20 and CD274 for mycoplasma contamination. No DNA band of approximately 502–520 bp, specific to mycoplasma, was detected in the newly constructed cell line (Figure 2G).

CAR-E276 exhibits significant anti-tumor growth activity in NSCLC CDX model

When the average tumor volume in the CDX model animals reached approximately 100 mm³, the animals were randomly assigned to five different treatment groups for the *in vivo* efficacy study (Figure 3A). To evaluate the *in vivo* anti-tumor efficacy of CAR-E276, we measured the tumor volume and recorded the

volume changes every 2–3 days during the experiment. After six treatments, all treatment groups exhibited slower tumor growth compared to the control group. Notably, the tumor growth in groups treated with CAR-E276 was significantly inhibited compared to the corresponding mock T groups; the vehicle group demonstrated the fastest tumor growth. Furthermore, in treated groups with CAR-E276, the inhibitory effect showed a dose-dependent correlation (Figure 3B).

Both the high-dose and low-dose CAR- $\gamma\delta$ T (CAR-E276) groups showed significantly stronger anti-tumor effects than the mock T group, as measured by

tumor growth inhibition rates based on absolute tumor volume compared to vehicle group (Figure 3C; Table S1). Notably, the anti-tumor efficacy of the CAR- $\gamma\delta$ T high-dose group was significantly greater than that of the mock T ($\gamma\delta$ T) high-dose group. Additionally, there was a significant dose-dependent difference in anti-tumor efficacy between the CAR- $\gamma\delta$ T high-dose group and the CAR- $\gamma\delta$ T low-dose group. These data indicated that the CAR structure significantly enhances the *in vivo* killing activity of $\gamma\delta$ T cells against tumors.

Due to the expression of luciferase in the A549-CD274-CCL20-Luc tumor cell line used to establish the NSCLC-CDX model, *in vivo* bioluminescence imaging was performed on Day 0, Day 7, Day 15, Day 22, Day 29, and Day 43 near the end of the study. Tumor growth *in vivo* was monitored based on changes in fluorescence intensity (Figures 3E–3G), which also

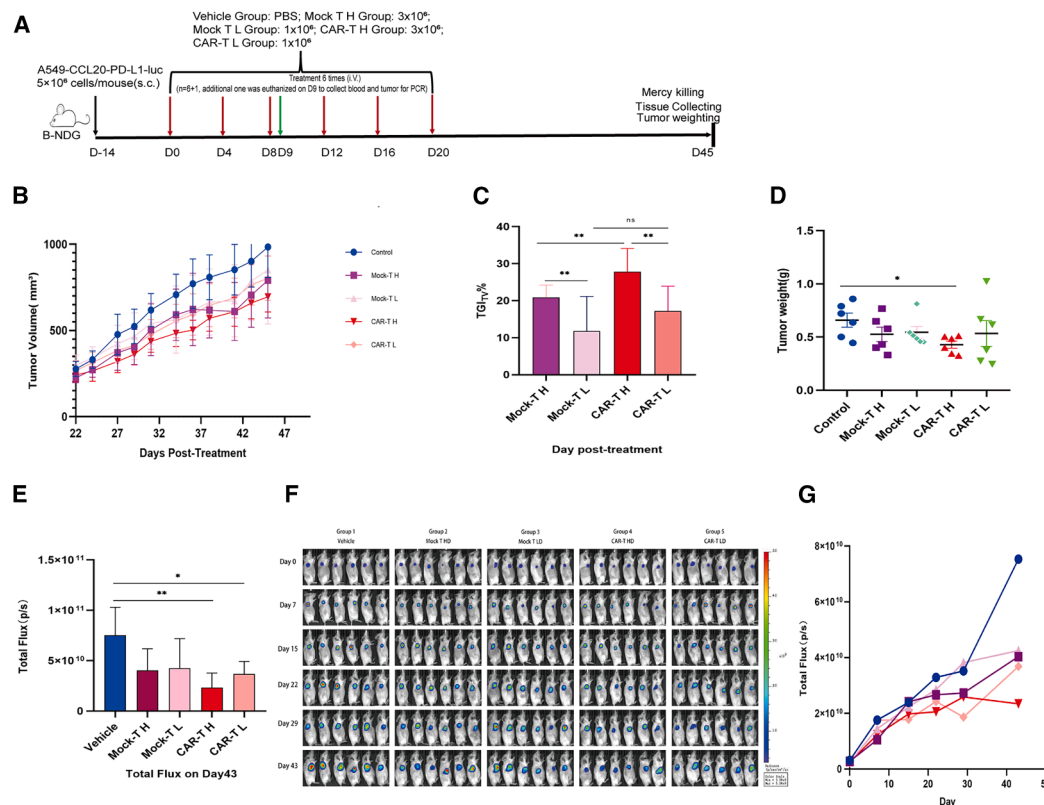


Figure 3. CAR-E276 exhibits significant *in vivo* efficacy in NSCLC CDX model

(A) Study schematic of *in vivo* efficacy for CAR- $\gamma\delta$ T in NSCLC CDX model.

(B) Tumor volume change curve of each group after administration. Data are presented as mean $\pm$ SD ($n = 6$, animals per group).

(C) Tumor growth inhibition rates of each treated group based on tumor volume change after treatment. The tumor growth inhibition rate was typically calculated after each measurement of tumor volume for animals ($n = 6$, animals per group) using the formula: $(1 - TV_T/TV_C) \times 100\%$, where TV_C denotes the average tumor volume in the control group and TV_T denotes the average tumor volume in the treatment group. A total of 11 independent measurements were performed, so the data are presented as mean $\pm$ SD of ($n = 11$, independent measurements). See also Table S1 for statistical power, effect size, 95% Confidence interval (CI) and specific p values.

(D) Tumor weight differences across groups. Data are presented as mean $\pm$ SD ($n = 6$, animals per group).

(E) Total flux (TF) value differences across groups on Day 43 near the end of the *in vivo* study. Data are presented as mean $\pm$ SD ($n = 6$, animals per group).

(F) *In vivo* imaging photographs of all animals ($n = 6$) in each group.

(G) Tumor total flux (TF) change curve in mice. Data are presented as the mean ($n = 6$, animals per group). Statistical analyses were performed using the two-tailed unpaired Student's t test. ns, not significant; * $p < 0.05$, ** $p < 0.01$.

demonstrated the significant inhibitory effect of the CAR- $\gamma\delta$ T on tumor growth *in vivo*.

Following the completion of treatment, we monitored the animals for an additional 25 days. On day 45, tumor-bearing B-NDG mice were euthanized, and tumor weights were measured across all groups. Based on the tumor weight data, we further calculated and compared the tumor growth inhibition rates; compared with the vehicle group, CAR-E276 demonstrated statistically significant tumor growth inhibition (Figure 3D). In summary, the assessment parameters, including changes in tumor volume, tumor weight, and *in vivo* bioluminescence imaging, collectively indicated that CAR-E276 exhibits significant *in vivo* anti-tumor efficacy.

CAR-E276 is effectively distributed to the tumor site *in vivo* and exhibits sustained persistence

In this *in vivo* efficacy study, whole blood and tumor tissue samples were collected from mice on Day 9 and Day 45. Genomic

DNA was extracted and PCR testing was performed using EGFR scFv-specific primers to assess the presence of CAR-E276 in blood and its distribution in tumor tissue. PCR products were analyzed by agarose gel electrophoresis. Only the high- and low-dose CAR-E276 groups were expected to display a target band of approximately 500 bp, whereas the mock T group (lacking a CAR structure) and the vehicle group (treated with PBS only) were not expected to yield an amplification band. The intended PCR products was successfully observed in both Day 9 (Figure 4A) and Day 45 (Figure 4C) blood samples, as well as in tumor tissue samples on Day 9 (Figure 4B) and Day 45 (Figure 4D), from both the high- and low-dose CAR-E276 groups. Overall, compared to blood samples collected at the same time points, the PCR product bands from tumor samples were relatively weaker, suggesting that only a subset of T cells infiltrated the tumor microenvironment, a finding consistent with biological expectations. In summary, the PCR results from blood and tumor

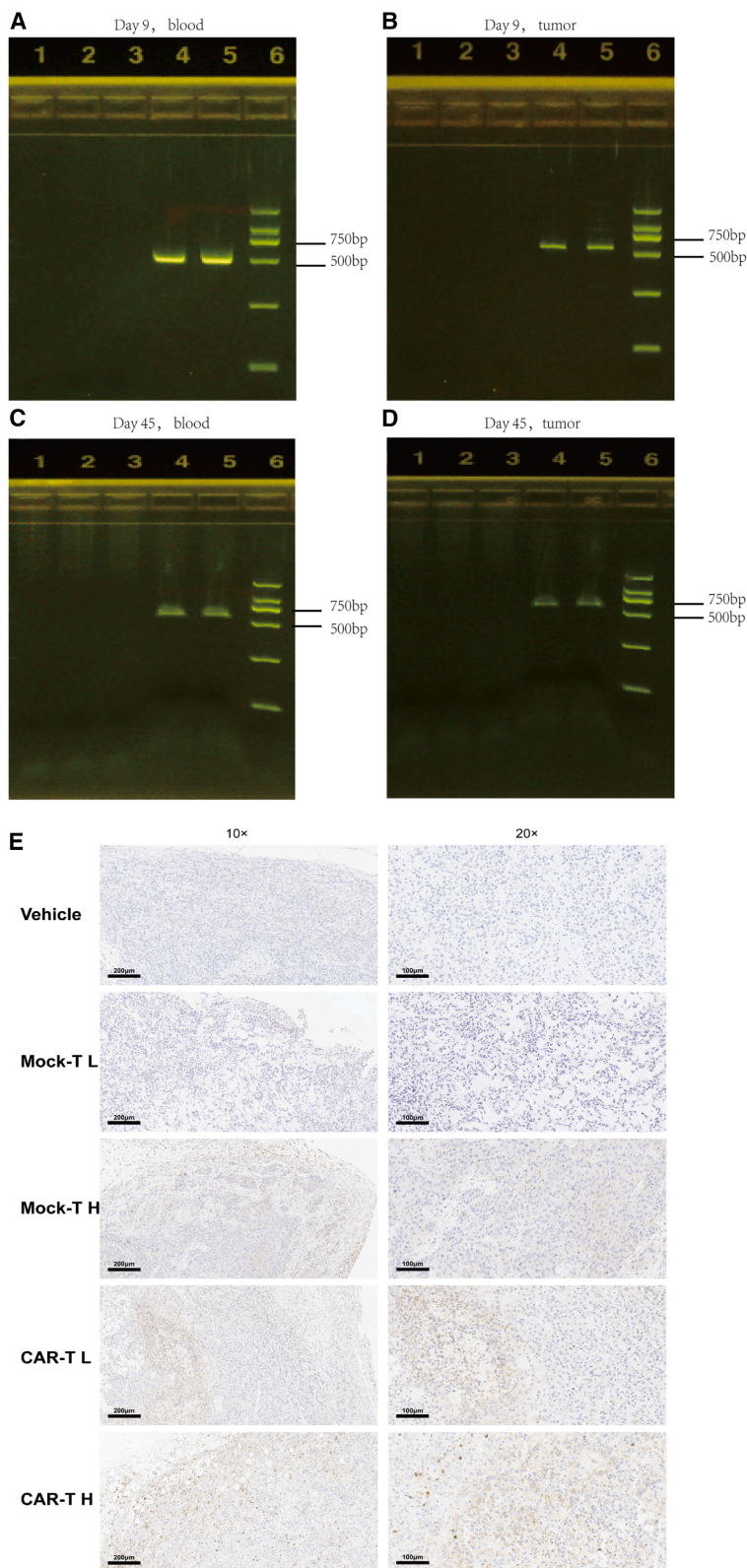


Figure 4. CAR-E276 demonstrates effective distribution, infiltration, and persistence *in vivo*

(A and B) PCR detection of CAR- $\gamma\delta$ T (CAR-E276) in blood and tumor tissues collected on the day of experiment termination (Day 9).

(C and D) PCR detection of CAR- $\gamma\delta$ T (CAR-E276) in blood and tumor tissues collected on the day of experiment termination (Day 45). In each agarose gel electrophoresis image, the samples in lanes 1 through 5 correspond to the vehicle group, high-dose mock T group, low-dose mock T group, high-dose CAR- $\gamma\delta$ T (CAR-E276) group, and low-dose CAR- $\gamma\delta$ T (CAR-E276) group, respectively.

(E) IHC was employed to identify the distribution of CAR-E276 in blood and tumor tissue. The yellow-stained particles represent $\gamma\delta$ T cells labeled with CD3 antibodies. A 10× field of view covers the tumor periphery extending toward the interior, while a 20× field of view captures regions closer to the tumor core. Scale bars, 200 μ m and 100 μ m.

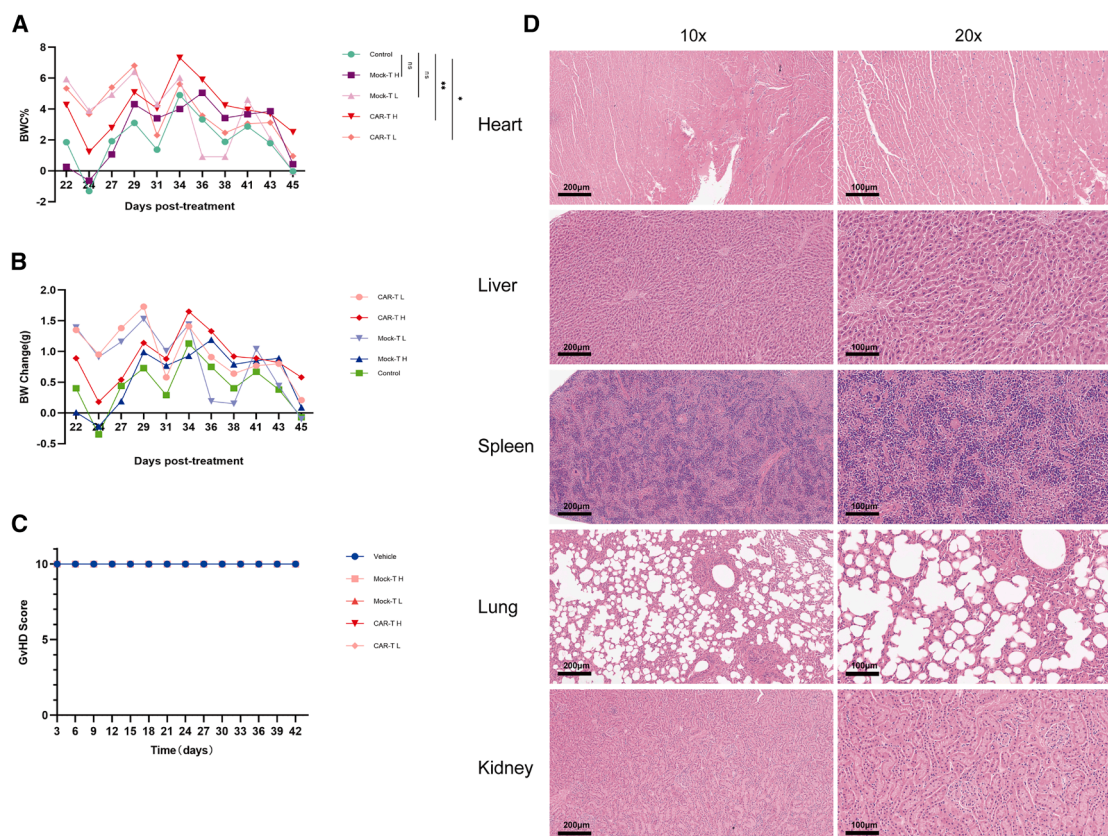


Figure 5. CAR-E276 exhibits favorable safety profiles and improves the growth condition of mice

(A and B) shows the curves for weight change rate (BWC%) and absolute body weight change (BW Change) of animals, respectively. The curves were fitted based on a total of 11 independent measurements after treatment. Data are presented as the mean ($n = 6$, animals per group).

(C) GvHD score. Data are presented as the mean ($n = 6$, animals per group).

(D) H&E staining results of major organs in mice. The paraffin tissue sections stained with H&E were collected at the end of the experiment on Day 45 (10 $\times$ and 20 $\times$ magnification), showing clear and intact tissue morphology with no observed pathological damage to major organs. Scale bars, 200 μ m and 100 μ m.

Statistical analyses were performed using the two-tailed unpaired Student's t-test. ns, not significant; * $p < 0.05$, ** $p < 0.01$.

samples confirmed that CAR-E276 can migrate into tumor tissues, facilitating its anti-tumor activity. Moreover, CAR-T cells remained detectable on Day 45, strongly indicating that this CAR-T demonstrates robust *in vivo* persistence. Based on these findings, to further evaluate the *in vivo* persistence of CAR-E276 and obtain preliminary pharmacokinetic data, we conducted an independent *in vivo* study (Figure S1A). The results from a typical qPCR-based PK assay method commonly used in the industry demonstrated that CAR-E276 remained detectable in blood samples collected on Day 71 (Figures S1B and S1C), indicating its prolonged persistence profile.

To further confirm that CAR-E276 can be transported to tumor site and infiltrate into the tumor tissues, we performed immunohistochemistry (IHC) using an hCD3 antibody to detect T cell infiltration (Figure 4E). Both the high- and low-dose groups of CAR-E276 exhibited significant T cell distribution, not only in the periphery but also within the tumor. Notably, the number of T cells in the high-dose group was significantly higher than that in the low-dose group. A certain amount of T cells was also observed in the periphery of the tumor in the mock-T group, which may be attributed to the inherent tissue tropism of $\gamma\delta$

T cells. However, no obvious T cell infiltration was detected in the mock-T group extending toward the interior of the tumor, indicating that CCR6 plays a critical role in promoting CAR-T chemotaxis and infiltration into the tumor microenvironment. These IHC results further demonstrated that the tumor inhibition efficacy of CAR-E276 is closely associated with its dose-dependent chemotactic infiltration into tumor tissue.

CAR-E276 promotes animal growth without GvHD and demonstrates excellent safety

In this study, we measured the body weight of the mice every 2–3 days and calculated the changes in body weight. The high- and low-dose groups of CAR-E276 exhibited significant advantages in body weight gain compared to the control group (vehicle group), while the two mock-T dose groups did not show significant differences in body weight gain when compared to the control group (Figures 5A and 5B).

To evaluate whether the injected effector cells would induce GvHD and to assess the potential of this CAR- $\gamma\delta$ T as an allogeneic, universal, off-the-shelf CAR-T therapy, we scored five parameters (weight, posture, activity, hair condition, and skin

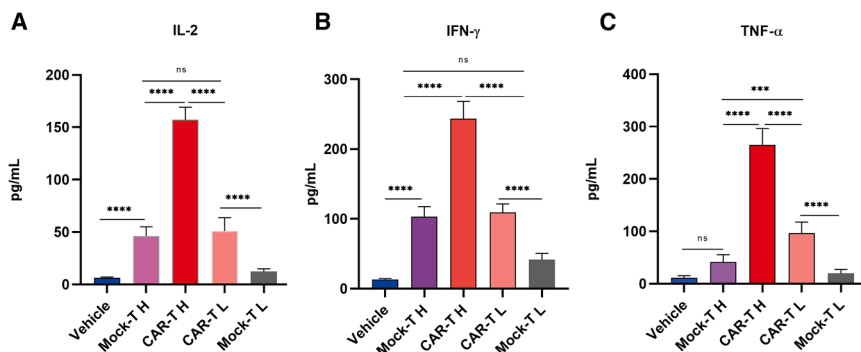


Figure 6. CAR-E276 releases cytokines *in vivo* that are critical for the execution of its activity

(A–C) ELISA analysis for the release of IL-2, IFN- γ , and TNF- α by CAR-E276 *in vivo*. Data are presented as mean $\pm$ standard deviation (SD) ($n = 6$, animals per group). Statistical analyses were performed using one-way ANOVA. ns, not significant; *** $p < 0.001$, **** $p < 0.0001$.

condition) on a scale of 0–2 points each, with lower scores indicating more severe GvHD symptoms. The total score for the five parameters was 10. After continuous monitoring throughout the study, no GvHD-related symptoms were observed in any of the treatment groups (Figure 5C).

The toxic side effects are typically observed at high doses. Therefore, at the conclusion of this study, we collected the heart, liver, spleen, lungs, and kidneys from the high-dose CAR-E276 treated mice and prepared paraffin-embedded sections for H&E staining. No pathological changes were observed in any of the organs, indicating that our CAR- $\gamma\delta$ T (CAR-E276) exhibits excellent safety (Figure 5D).

In this study, preliminary safety evaluations were conducted on tumor-bearing B-NDG mice treated with CAR- $\gamma\delta$ T cells through multiple aspects, including weight measurement, animal observation, and pathological testing. Overall, no manifestations related to graft-versus-host disease (GvHD), significant weight loss, or lesions in major organs were observed. These findings provided preliminary evidence suggesting that $\gamma\delta$ T-based CAR-T cells are indeed relatively safe due to $\gamma\delta$ T's inherent characteristics. Additionally, CAR-E276 also exhibited good safety profiles and caused no significant toxic side effects, thereby improving the overall health status of the animals.

CAR-E276 exhibits robust *in vivo* cytokine release capability

CAR- $\gamma\delta$ T can exert its activity and killing effects by releasing cytokines such as TNF- α , IL-2, and IFN- γ ; therefore, the killing process may be accompanied by the release of these cytokines. During this study, plasma samples were collected from B-NDG tumor-bearing mice on Day 3, Day 7, and Day 11 post-administration. The secretion levels of cytokines, including TNF- α , IL-2, and IFN- γ , in the plasma samples were quantified using ELISA. The overall cytokine release level of CAR-E276 was significantly higher than that of unmodified $\gamma\delta$ T cells (mock-T). Furthermore, the level of cytokines released by CAR-E276 showed a dose-dependent relationship (Figures 6A–6C).

DISCUSSION

It is estimated that there are 2 million new cases of lung cancer and 1.8 million deaths annually.^{24,25} Lung cancer is currently the most prevalent cancer in China.²⁶ Based on histopathology, lung cancer is basically divided into NSCLC and SCLC, with

NSCLC accounting for the vast majority, about 85% of all lung cancer cases.²⁷

Therefore, finding more effective or innovative treatment strategies for lung cancer, especially NSCLC, remains a key focus for many researchers and biopharmaceutical companies.

Currently, multiple small molecule drugs targeting genes such as EGFR, ROS1, and ALK have been developed for the treatment of various subtypes of NSCLC patients. Among them, EGFR is a prominent target for NSCLC therapy. Although targeted therapy drugs have significantly improved the clinical outcomes of a considerable proportion of advanced NSCLC patients, they still face challenges in extending progression-free survival (PFS) and overall survival (OS). Additionally, their combination with monoclonal antibodies has not yet fully overcome these limitations, along with the inevitable issue of drug resistance.^{28,29} In recent years, immune checkpoint inhibitors and immune cell therapies have emerged as the two primary directions for advancing tumor immunotherapy globally.³⁰ Given that any single therapy among the two major immunotherapy approaches has potential limitations, the integration of these or even additional approaches represents a promising strategy. CAR-T serves as an optimal platform for implementing this integrated therapeutic strategy. It enables the integration of multiple therapeutic modalities through genetic engineering by utilizing immune cells as functional carriers. This approach may reduce the operational complexity and cost associated with combination therapy during clinical treatment. The CAR-E276 we are reporting here is an integrated CAR- $\gamma\delta$ T against EGFR, combining immune checkpoint inhibitors with immune cell therapy.

The limited therapeutic efficacy of CAR-T in solid tumors, including NSCLC, can primarily be attributed to several factors, such as insufficient CAR-T cell trafficking, delivery, and infiltration into tumor tissue, as well as the immunosuppressive tumor microenvironment.^{31,32} Emil R. Bulatov's group also provided further evidence suggesting that the overall low efficacy of CAR-T therapy against solid tumors might be conditioned by such tumor features as dense stroma or immunosuppressive microenvironment.³³ The inefficient transport of CAR-T cells to solid tumor sites represents a significant obstacle to the efficacy of CAR-T therapy. In addition to extracellular matrix and cellular physical barriers, this limitation may also stem from the mismatch between chemokines secreted by tumors and chemokine receptors on CAR-T cells, or the absence of appropriate receptors on effector CAR-T lymphocytes targeting low-chemokine-producing tumor cells.^{34–36}

Chemokines work in concert with their receptors on the cell surface to guide cells to specific locations throughout the body and induce directional migration. Chemokine molecules and their receptors play crucial roles in regulating cell entry and exit from the tumor microenvironment and are key participants in cancer biology.³⁷ Leveraging chemokine networks for tumor therapy represents a promising strategy.³⁸ Since the discovery of chemokines CXCL8 (IL-8) and CCL2 (MCP-1) in the late 1980s, the members of the chemokine superfamily have rapidly expanded,³⁹ with C-C chemokine ligand 20 (CCL20) and C-C chemokine receptor 6 (CCR6) being notable members. CCL20 is a small protein expressed in various tissues and cells, including the liver, lung, colon, and skin. It plays a role in tissue inflammation and homeostasis and specifically interacts with its receptor, CCR6. CCR6 is predominantly expressed on diverse immune cells, such as B cells, immature dendritic cells, innate lymphoid cells (ILCs), regulatory CD4 T cells, and Th17 cells.⁴⁰ Based on these findings, upregulating CCR6 expression in CAR-T cells is a promising strategy.

Researches indicate that CCL20 is highly expressed in numerous solid tumors, and the CCL20-CCR6 axis is associated with various cancers, including lung cancer, hepatocellular carcinoma, colorectal cancer, breast cancer, pancreatic cancer, cervical cancer, and kidney cancer.^{41,42} A study by Fan et al. demonstrated that high CCL20 expression in lung adenocarcinoma serves as a poor prognostic marker for patients with this type of lung cancer.⁴³ A series of research findings suggested that CCR6 and CCL20 may play roles in lung adenocarcinoma, promoting proliferation and migration via autocrine or paracrine mechanisms.⁴⁴ Jin et al.⁴⁵ found that CCL20 is highly expressed in lung cancer and other high-incidence or high-mortality cancers, their research showed that the expression of CCR6, a specific receptor for CCL20, can robustly direct CAR-T cells toward tumor cells secreting CCL20. CAR-T cells expressing CCR6 can effectively migrate to and infiltrate solid tumors after injection, achieving efficient tumor clearance and significantly prolonging the survival time of xenograft tumor-bearing mice. Although studies have suggested that CCR2- or CXCR2-engineered CAR-T cells tend to migrate to lung tumors,⁴⁶ a major concern arises from the fact that these receptors can be activated by multiple ligands highly expressed in normal tissues, potentially leading to severe toxicity to normal organs. In contrast, CCR6 is uniquely known as the sole receptor for CCL20 and shows significantly higher expression levels in lung tumors compared to normal tissues. Notably, many chemokines exhibit promiscuous binding properties, interacting with multiple receptors, while receptors themselves can bind to various ligands. Therefore, the highly selective interaction between CCL20 and CCR6 is a focal point of our utilization in this research.⁴⁷ This specificity promotes tumor infiltration by CAR-T cells in CCL20-secreting environment.⁴⁸ The other key rationale for this CAR-T (CAR-E276) to selectively express CCR6 is to minimize off-target toxicity.

PD-L1 is expressed on tumor cells, and tumors can evade immune surveillance by exploiting the inhibitory immune checkpoint pathway involving the binding of PD1 and its ligand PD-L1. Antibody drugs targeting PD1 or its ligand PD-L1 have significantly transformed the treatment landscape of NSCLC. Consequently, CAR-T cells targeting immune checkpoints represent

a promising strategy for NSCLC and have demonstrated encouraging effects.⁴⁹ The strategy of modifying CAR-T cells by interfering with PD1 to counteract immunosuppressive mechanisms in the tumor microenvironment has been explored by some scientists.^{50,51} For instance, the cytoplasmic expression of PD1-CD28 chimeric structures on CD28-expressing cells can convert inhibitory signals into activating signals, while co-administration of PD 1 antibodies with CAR-T cells or genetic knockout of PD1 can enhance anti-tumor immune responses.^{22,52,53} However, these methods can be quite cumbersome to apply in clinical practice. Although the combination of PD1 or PD-L1 antibodies with CAR-T therapy has been reported to demonstrate efficacy in cancer treatment, this approach presents challenges in clinical practice due to its complexity, involvement of multiple drug types and administration schedules, and lack of cost reduction. Furthermore, the therapeutic efficacy of antibody combinations may vary depending on the pharmaceutical source. The concurrent secretion of immune checkpoint inhibitors, such as anti-PD1 scFv, by CAR-T cells (e.g., CAR-E276) offers a more streamlined strategy to address these limitations.

Incorporating the immune checkpoint roadblock-clearing function and the chemokine-guided traction function into $\gamma\delta$ T cells, which lack MHC restriction and exhibit better tissue tropism, via CAR-E276 is an ideal strategy that avoids increasing the complexity and cost of the manufacturing process. $\gamma\delta$ T cells utilization can be helpful to control CRS and other adverse events resulting from CAR T cell therapy. V δ 2 T cells are the main subgroup we used for CAR-E276 development, and they are derived from human peripheral blood and also the main $\gamma\delta$ T cells in the peripheral blood.^{54,55} V δ 2 T cells are a versatile subset for cellular immunotherapy, having many helpful properties that include potent antibody dependent cellular cytotoxicity capacity, a tissue tropic homing profile, and a range of innate tumor-sensing receptors minimizing the likelihood of tumor escape.⁵⁶ Natural receptors enable these cells to circumvent the conventional resistance mechanisms associated with traditional chemotherapeutic agents. Due to the MHC-independent characteristic of $\gamma\delta$ T cells, tumor cells cannot be resistant to immunotherapy by downregulating MHC expression. Drug resistance remains a challenge for CAR-T immunotherapy. Drug resistance remains a major challenge in CAR-T immunotherapy. However, these features highlight the advantage of utilizing $\gamma\delta$ T cells as source cells over $\alpha\beta$ -T cells, as they possess superior capacity to overcome drug resistance. Certainly, there are many mechanisms underlying CAR-T resistance, including antigen escape, immunosuppression, and CAR-T cell dysfunction or exhaustion.^{57,58} The prevailing strategy involves utilizing dual or multi-targeted CARs to prevent antigen escape and inhibit immune-suppressive pathways, such as PD1/PD-L1, thereby overcoming the dysfunction and exhaustion of immune cells. For CAR-E276, in addition to constructing EGFR-targeted CARs based on $\gamma\delta$ T cells, it also secretes anti-PD1 scFv. This characteristic enables CAR-E276 cells to function as dual- or multi-targeted CAR-T cells, thereby enhancing their potential to overcome resistance that may develop during future clinical applications.

In the *in vitro* killing effect experiments, the co-culture of effector cells and target cells often leads to direct contact, which

easily triggers the killing effect. Therefore, the *in vitro* efficacy of CAR-T is merely a conceptual functional characterization. The *in vivo* processes of CAR-T therapy are inherently complex, encompassing delivery, migration, infiltration, cytotoxic activity, persistence, and cytokine release, need to be validated and elucidated using appropriate animal models that can fully reflect the mechanism of CAR-T action. This *in vivo* study using a CDX model showed that CAR-E276 exhibited significant anti-tumor efficacy. Biodistribution evaluation using PCR and IHC in this study demonstrated that intravenously injected CAR-E276 cells were successfully chemotactically guided and infiltrated into tumor tissue. ELISA assay confirmed that CAR-E276 can effectively release TNF- α , IFN- γ , and IL-2 *in vivo*, indicating that it exerts an indirect killing effect through the release of cytokines. Based on daily animal observations and weight measurements, no abnormalities were observed in the skin, hair, posture or weight gain of mice. Further pathological examination revealed no signs of GvHD, confirming that $\gamma\delta$ T-based CAR-T does not cause GvHD as theoretically expected.

In order to comprehensively investigate and demonstrate the promoting role of the CCL20-CCR6 axis in the migration and infiltration of CAR-E276 into tumor tissues, we employed intravenous injection via the mouse tail vein, which fully illustrated the chemotactic function of CCR6 in CAR-E276. When characterizing the chemotactic function, we utilized subcutaneous tumor modeling rather than intraperitoneal tumor modeling. Some studies used intraperitoneal modeling combined with intraperitoneal drug injection, or intratumoral injection was employed directly.^{54–59} Such tumor-bearing mice models and the administration route of effector cells may exaggerate the efficacy of CAR-T, potentially deviating from clinical practice and compromising the objectivity of drug efficacy evaluations. Moreover, in clinical practice, the administration of effector cells via intraperitoneal or intratumoral injection is technically challenging and may impose additional psychological burdens on patients. Currently approved CAR-T therapies, as well as TCR-T therapies, are all administered via intravenous injection. In the CAR-E276 study, we fully considered the specific scenarios of actual clinical applications.

Secondly, during modeling, we employed a higher dose of tumor cells rather than the lower dose of 3×10^6 cells/mouse used in our previous study on CAR- $\alpha\beta$ T,⁶⁰ where effector cells were administered only at the onset of tumor formation. The focus of this study was on application, aiming to align as closely as possible with future clinical applications in the biopharmaceutical industry. In this study, we modeled animals with a higher dose of 5×10^6 tumor cells/mouse and administered effector cells under conditions of high tumor burden, with a tumor volume of 100 mm³. Due to the late onset of symptoms in lung cancer, many patients are already in advanced stages once diagnosed with lung cancer.³¹ Therefore, the dosing regimen in this study more closely reflected the real-world scenario at the time of diagnosis for this type of cancer. Here, we emphasize that the use of animal models and *in vivo* drug administration should not solely aim for striking results but should be value-oriented, fully reflecting the drug design concept and being closely tied to actual clinical applications. Indeed, only 5% of anticancer drugs demonstrating activity in preclinical studies were ultimately approved

for clinical use by the United States FDA,^{61–63} highlighting the critical importance of selecting appropriate models for preclinical drug evaluation. In addition to animal models, whether the administration route can objectively reflect results and clinical applicability is also critical. Based on this CDX model, our study design was fully integrated with clinical application scenarios, and the data demonstrated that CAR-E276 exhibits significant anti-tumor efficacy even under high tumor load. Moreover, our dose was very low temporarily, with a very wide dose-adjustment window, indicating potential for the CAR-T field to progress toward first-line treatment. Whether CAR-T can advance to first-line treatment in the future remains an industry concern.

Once more, a key challenge of CAR-T cell therapy lies in depletion and persistence, which may impact its long-term effectiveness. Unlike previous studies where animals were euthanized immediately after several rounds of drug administration,⁵⁹ which did not facilitate observation of sustained effects, judging tumor growth inhibition rates shortly after initial administration was relatively one-sided. Because the killing ability of T cells may be pronounced initially, while sustained changes in T cell depletion or viability have yet to manifest. A short observation period is not conducive to comprehensive evaluation. For a more thorough and objective investigation, our study maintained an observation period of over two weeks following the completion of all drug treatments whenever possible. Our results indicated that CAR-E276 exhibits excellent persistence *in vivo*, which served as the foundation for its significant *in vivo* tumor inhibition.

In summary, CAR-E276 is a CAR-T therapy product for NSCLC that leverages the tissue tropism of $\gamma\delta$ T cells and the specific chemotaxis of the CCL20-CCR6 axis to enhance CAR-T delivery, infiltration, and efficacy at the tumor site. It achieves this by secreting PD1 blockades to overcome immunosuppressive barriers in the tumor microenvironment and promote the persistence of effector cells *in vivo*. Based on the core design principles of this CAR-T therapy, we have established an NSCLC-CDX model adhering to industry modeling standards to comprehensively validate and evaluate the *in vivo* efficacy of CAR-E276. Our study design fully aligned with its mechanism of action and clinical application scenarios in NSCLC. The data objectively reflected the effectiveness of current CAR-T therapies for solid tumors. The results demonstrated that the CAR-T design concept exemplified by CAR-E276, which integrates chemokine-guided trafficking and immune checkpoint inhibition while utilizing MHC-independent $\gamma\delta$ T cells as seed cells, represents a highly promising strategy for the treatment of solid tumors. This approach also offered a valuable pathway toward off-the-shelf and allogeneic universal CAR-T therapies.

Limitations of the study

This study represents a preliminary exploratory proof-of-concept investigation. First, the CAR-E276 validated in this study primarily targets NSCLC patients who are triple-positive for EGFR (excluding extracellular domain mutations), CCL20, and PD-L1. Clinical application requires prior genetic screening of patients. However, tumors are generally heterogeneous in clinical settings; even among patients positive for all three markers, expression

levels may vary significantly, which differs from the tumor-bearing mouse models derived from single-source tumor cell lines. Furthermore, the specific threshold levels of these target indicators for effective CAR-T therapy remain to be determined and should be investigated in future studies. In addition, although $\gamma\delta$ T cells are highlighted for their MHC-independent activation and reduced cytokine release, these characteristics also raise certain concerns. On one hand, non-MHC-restricted immune cells facilitate the development of allogeneic universal therapies but may lead to indiscriminate cytotoxicity. On the other hand, the generally lower cytokine release compared to $\alpha\beta$ T cells¹⁶ may help mitigate the risk of CRS, but could also compromise their overall immune regulatory efficacy within the tumor microenvironment and their synergistic interactions with other immune cell populations. These concerns are not absolute but are dose-dependent, necessitating careful balancing with appropriate dosage regimens in future research. Moreover, in this study, the safety profile of CAR-E276 was assessed only through animal behavior, weight changes, and histopathological examination of major organs. The immunogenicity of the CAR construct, the tumorigenic potential of the effector cells, and potential hematotoxic effects were not evaluated. Due to limitations in sample collection and the lifespan of the immune-deficient mouse model, these aspects were not included in the current study. To support clinical translation, future studies should incorporate dedicated toxicology and pharmacokinetic distribution assessments, which will require additional financial support to ensure a more comprehensive evaluation of the therapy's safety and efficacy.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Dengji Zhang (zhdengji518@163.com).

Materials availability

This study did not generate new unique reagents.

Data and code availability

This paper does not report original code. All data associated with this study are present in the paper or the supplemental information. Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, H.Z. and X.Y.; methodology, D.Z. and X.Y.; cell experiments and all data analysis, D.Z., X.Y., Y.T., Y.C., Y.Z., and X.Z.; animal experiments, D.Z., W.S., Y.G., and Y.C.; writing—original draft, D.Z. and Y.T.; writing—review and editing, D.Z., Y.T., Y.Z, X.Z., and X.Y.; funding acquisition, H.Z. and D.Z.; resources, H.Z., D.Z. and X.Y.; supervision, H.Z. and X.Y. All authors have read and agreed to the published version of the article.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-hB7-H1 Alexa Fluor conjugated mouse IgG1	R&D Systems	Cat#: FAB1561G RRID: AB_10994182
Human EGFR PE-conjugated Antibody	R&D Systems	Cat#: FAB9577P RRID: AB_2942015
CD3 epsilon Recombinant Rabbit mAb	STARTER	Cat#: S0B2132
Anti-human LAG3	R&D Systems	Cat#: BAF2319 RRID: AB_416736
Anti-human Tim3	R&D Systems	Cat#: BAF2365 RRID: AB_2279579
Anti-human CD279	R&D Systems	Cat#: BAF 1086 RRID: AB_355833
Bacterial and virus strains		
DH5α	TIANGEN	Cat#: CB101-03
psPAX2	Addgene	Cat#: 12260
pMD2G	Addgene	Cat#: 12259
lentiviral plasmid vectors FUGW-CCL20-T2A-hgro	Fudan	N/A
pcdh-CMV-LUC-EF1A-CD274-PA2-PURO	Fudan	N/A
Chemicals, peptides, and recombinant proteins		
Matrigel	Corning	Cat#: 354234
XenoLight D-Luciferin	PerkinElmer	Cat#: 122799
Isoflurane	RWD lifescience	Cat#: R510-22-10
FBS	GIBCO	Cat#: 10091-148
Trypsin	GIBCO	Cat#: 25200072
PBS	GIBCO	Cat#: 10010001
F12K	GIBCO	Cat#: 21127030
DMEM	GIBCO	GIBCO
Plasmid Extraction Kit	Macherey-Nagel	Cat#: 740412
ReverTra Ace® qPCR RT Kit	TOYOBO	Cat#: FSQ-301
SYBR Green Master Mix	TOYOBO	Cat#: QPK-201
Lipofiter™ transfection reagents	Hanbio	Cat#: HB-TRCF-1000
EcoRI	Thermo	Cat#: ER0271
BamHI	Thermo	Cat#: ER0052
APC Streptavidin	BD Bioscience	Cat#: 554067
T-PER Tissue Protein Extraction Reagent	Thermo	Cat#: 78510
Gel DNA recovery reagent	Generay	Cat#: GK2041
DNA ladder	Generay	Cat#: GsDL10004-250
E-Gel®EX 1% Agarose	Invitrogen	Cat#: G401001
Trans2K DNA Marker	TransGen	Cat#: BM101
PureLink® DNA Mini Kit	Thermo	Cat#: K1820-00
MaxVision™ HRP IHC Kit	MXB	Cat#: KIT-5005
Critical commercial assays		
Mycoplasma assay kit	Hanbio	Cat#: Myco-PCR-TEST
LEGEND MAX™ Human CCL20 ELISA Kit	Biolegend	Cat#: 441407
Human IL-2 ELISA Kit	MultiSciences	Cat#: 70-EK102-96
Human IFN-γ ELISA Kit	MultiSciences	Cat#: 70-EK180-96

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Human TNF- α ELISA Kit	MultiSciences	Cat#: 70-EK182-96
ONE-Glo™ EX Luciferase Assay System	Promega	Cat#:E8110
Experimental models: Cell lines		
A549	ATCC	Cat#: CCL-185 RRID: CVC_L_0023
HERK-293T	Medicilon	N/A
Experimental models: Organisms/strains		
B-NDG Mice (NOD.CB17-Prkdcscid Il2rgtm1Bcgen/Bcgen)	BIOCYTOGEN	Product#:110586
Oligonucleotides		
RT-qPCR primers for detection of overexpression of transgenes CD274 and CCL20, See Table S2 in supplemental information	This Paper	N/A
Primers of CAR for biodistribution, See Table S3 in supplemental information	This Paper	N/A
Software and algorithms		
GraphPad Prism software (version 8.0.1)	GraphPad	N/A
IBM SPSS Statistics (Version 27)	IBM	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**Animals**

6-8 weeks old female B-NDG (NOD.CB17-Prkdcscid Il2rgtm1/Bcgen) mice with body weight of 16-18g were purchased from Bio-cytogen Pharmaceuticals Co., Ltd (Beijing, China), were housed in the SPF Laboratory Animal Center at medicilon. B-NDG mice were housed in IVC cages, with 3 mice per cage, in SPF grade animal rooms under 12-h light-dark cycles, with humidity between 30% and 70% at an ambient temperature of 20–26°C. They had free access to irradiated feed and drank filtered and high-pressure sterilized tap water. The animal study in this article received approval from the Institutional Animal Care and Use Committee of Shanghai Medicilon Inc (Approval number: MDT2310P). All the methods used in the study adhered to the relevant guidelines and regulations set forth by the committee. Before the experiment, the mice underwent at least one-week environmental adaptation period to reduce experimental stress, and a health assessment was conducted before study to confirm no signs of disease. Euthanasia was performed by carbon dioxide inhalation upon completion of the experiment or when the animal welfare termination criteria was met. The animal would be placed in euthanasia induction chamber, followed by the administration of carbon dioxide at a flow rate of 30% to 70% of the chamber volume per minute. Once the animal was fully anesthetized, it would be removed and subjected to cervical dislocation to ensure complete euthanasia.

When the average tumor volume of the non-small cell lung cancer-CDX model animals constructed using A549-CD274-CCL20-Luc tumor cells reached approximately 100 mm³, the animals were randomly assigned into five treatment groups: Control group (Vehicle group), Mock-T high-dose group (Mock-T H), Mock-T low-dose group (Mock-T L), CAR-E276 high-dose group (CAR-T H), and CAR-E276 low-dose group (CAR-T L). The number of animals per group is presented in the corresponding figure ([Figure 3A](#)). To ensure baseline comparability of tumor volumes across groups, the inter-group difference was controlled to be less than 10% of the mean during randomization. To avoid confounding factors, each cage was labeled with an experimental information card detailing mouse information and test article information, animal experiment details, and personnel information. Additionally, mice were individually marked using the ear-labeling method, and each animal was assigned a numerical identifier within its treatment group. All animals were included and received the appropriate dosage of the effector cells, Mock-T cells (unmodified $\gamma\delta$ T) or vehicle (PBS) according to their assigned treatment group. At the end of the experiment (Day 45), tumor-bearing mice were euthanized, and tumor and major organ tissues (e.g., heart, liver, skin, lungs, and kidneys) were collected for further analysis.

METHOD DETAILS**Construction of expression vectors**

Primer pairs for PCR amplification were designed based on the CD274 and CCL20 target gene sequences from the pcdh-CMV-LUC-EF1A-CD274-PA2-PURO plasmid and the FUGW-CCL20-T2A-hgro plasmid ([Table S2](#)). To facilitate subsequent cloning of the target genes CD274 and CCL20, Eco RI and Bam HI restriction enzyme recognition sites were incorporated during primer design. According to the manufacturer's instructions, the HB Infusion™ one-step cloning kit was employed to ligate the amplified target fragments

(CD274 and CCL20) into the enzyme-digested shuttle vector (pHBLV-CMV-MCS-EF1-fLUC-T2A-Puro) via homologous recombination technology. The resulting construct was designated as the expression plasmid vector LV-h-CD274-P2A-CCL20.

Packaging of lentiviruses

The HEK-293T cell line was transfected using Lipofiter™ transfection reagent (Hanbio Biotechnology, Shanghai, China). A three-plasmid system comprising PMD2.G, psPAX2, and the gene expression plasmid LV-h-CD274-P2A-CCL20 was used to produce lentivirus at a ratio of 1:2:2. After transfection, virus-containing culture supernatants were collected at 48- and 72-hours post-transfection. The collected supernatant was centrifuged at 2000g for 10 minutes at 4°C. The resulting supernatant was then transferred to high-speed centrifuge tubes and subjected to ultracentrifugation at 82,700g for 2 hours at 4°C. Following removal of the supernatant, the viral pellet was resuspended in complete culture medium, incubated overnight at 2–8°C, and subsequently transferred to sterile virus storage tubes. The concentrated virus was then stored in an ultra-low temperature freezer at below –60°C for future use.

Establishment of NSCLC cell line overexpressing CD274 and CCL20

A549 cells were seeded onto cell culture plates at a density of 1.5×10^5 cells/mL. On the following day, the multiplicity of infection (MOI) was determined, and the corresponding volume of virus stock was added based on an MOI of 1. Additionally, Polybrene co-infection reagent was included to achieve a final concentration of 7 µg/mL. The cells were monitored 48 hours post-transduction. When the cells reached 80–90% confluence, they were subcultured into 10 cm culture dishes for further expansion. Following two rounds of selection with puromycin, cells with high survival rates were expanded at appropriate ratios. As a result, a stable cell line designated A549-CD274-CCL20-Luc was successfully established.

Establishment of NSCLC CDX model

A549-CD274-CCL20-Luc human tumor cells were cultured in expanded flasks to achieve the required cell number for the planned animal experiments. Cells in the logarithmic growth phase were collected, washed three times with PBS, and resuspended in F12K basic medium containing 50% Matrigel. The cell concentration was adjusted to 5×10^7 cells/mL. Under sterile conditions, 0.1 mL of the cell suspension (containing 5×10^6 cells) was subcutaneously inoculated into the right flank of B-NDG mice at a dose of 5×10^6 cells/0.1 mL per mouse.

Test article

The test article, CAR-γδ T cells (referred to as CAR-E276), which express a chimeric antigen receptor targeting EGFR and the chemokine receptor CCR6, and secrete an scFv fragment of the anti-PD1 monoclonal antibody (Figure S2A), were constructed using standard lentiviral packaging protocols and provided by our laboratory at Fudan University. The exhaustion phenotypes of CAR-E276, compared with those of Mock-T cells, were assessed by flow cytometry, demonstrating a significantly lower exhaustion phenotype in CAR-E276 (Figure S2B). Furthermore, CAR-E276 exhibited superior performance compared to corresponding αβ CAR-T cells with identical genetic modifications, as evidenced by enhanced *in vitro* cytotoxic activity and increased cytokine release capacity (Figures S2C and S2D).

Luciferase assay

After culturing A549-CD274-CCL20-Luc cells and A549 Control cells for 4 hours in an opaque white culture plate, the plate was removed from the incubator and allowed to equilibrate to room temperature. The cell supernatant was then discarded, and 100 µL of Luciferase Assay System solution, prepared according to the ONE-Glo™ EX Luciferase Assay System kit manual (Promega, USA), was added to each well. Following a 3-minute lysis and reaction period at room temperature in the dark, luminescence signals were measured using a luminescence microplate reader.

Mycoplasma identification

After the newly established cell line was passaged, it was cultured in fresh F12K medium supplemented with 10% fetal bovine serum and 1.5 µg/mL puromycin for 48 hours. Subsequently, total RNA was extracted from the cell culture using Trizol reagent and reverse-transcribed for mycoplasma identification. Mycoplasma contamination was then assessed via PCR in accordance with the instructions provided in the mycoplasma assay kit manual (Hanbio Biotechnology, Shanghai, China).

FACS assays

Approximately 1×10^6 cells were collected and transferred into 5 mL flow cytometry tubes. The collected cells were centrifuged and washed with PBS and then incubated on ice for 30–45 minutes with 5–10 µL of corresponding fluorescein-labeled antibodies respectively. After antibody incubation, the cells were washed again with PBS, resuspended in 500 µL of pre-cooled PBS, filtered through a cell strainer, and analyzed using Beckman CytoFlex flow cytometry.

In vitro killing experiments

A549-CD274-CCL20-Luc cells at a density of 1×10^5 were seeded in 24-well plates. After 24 hours, different numbers of CAR-γδ T cells and CAR-αβ T cells (0 , 1.0×10^5 , 5.0×10^5 , and 1.0×10^6) were co-incubated with A549-CD274-CCL20-Luc cells for 8 hours,

respectively. The cytotoxicity of CAR- $\gamma\delta$ T cells and CAR- $\alpha\beta$ T cells against A549-CD274-CCL20-Luc cells was compared at different effector-to-target ratios by detecting the level of luciferase.

ELISA assay

Whole blood was collected from each mouse into anticoagulant tubes containing K₂EDTA. After being kept on ice for 30 minutes, the blood samples were centrifuged at 4°C and 2000g for 30 minutes to obtain plasma. The plasma samples were then placed on ice, transferred to a -80°C ultra-low temperature freezer, and stored until subsequent cytokine analysis. Cytokine levels, including IL-2, IFN- γ , and TNF- α , were measured using ELISA kits from MultiSciences Company, and all procedures were performed in accordance with the manufacturer's instructions.

The culture medium of the A549 stable cell line was collected and used for CCL20 detection. CCL20 levels were measured using the LEGEND MAX™ Human CCL20 ELISA Kit, and all procedures were performed strictly according to the manufacturer's instructions.

RT-qPCR detection of overexpression of transgenes

The newly constructed cell line was lysed with Trizol reagent, and total RNA was extracted. Reverse transcription was carried out using the ReverTra Ace® qPCR RT Kit in accordance with the manufacturer's instructions. Primer pairs designed based on the sequences of the human CD274 and human CCL20 genes were employed for the relative quantitative analysis of gene expression by the SYBR Green method, with hGAPDH used as the reference gene. All primer sequences are listed in the [supplemental information](#) section ([Table S2](#)).

PCR assay for biodistribution

Mouse whole blood was collected from the eye socket and anticoagulated with K₂EDTA for subsequent DNA extraction using the PureLink® DNA Mini Kit. A pair of primers designed based on the sequence of CAR targeting the EGFR region was used for PCR detection after DNA extraction ([supplemental information](#), [Table S3](#)). Tumor tissues were homogenized, and DNA was extracted following the protocol outlined in the PureLink® DNA Mini Kit manual (Thermo Fisher Scientific) for PCR analysis. The same PCR amplification program was applied to both blood and tissue samples.

GvHD observation and score

GvHD was observed and scored according to five factors: weight loss, body posture, motility, fur condition, and skin exposure. Each factor was scored on a scale from 0 to 2, with a lower score indicating more severe GvHD at the corresponding time point. The total score was calculated as the sum of the scores across all five factors.

Histological analysis

Following euthanasia, the heart, liver, spleen, lungs, kidneys, and tumor were excised. The tissues were fixed in 4% paraformaldehyde (PFA) and processed for paraffin sectioning. To assess the infiltration of effector cells into tumors, immunohistochemical analysis was carried out for tumor sections using an anti-hCD3 antibody. Hematoxylin and eosin (H&E) staining was performed on sections of the heart, liver, spleen, lungs, and kidneys to evaluate potential tissue damage. Both immunohistochemical and H&E analyses were examined under a microscope at 10× and 20× magnification.

In vivo bioluminescence live animal imaging (BLI)

15 minutes prior to imaging, 150 mg/kg of D-luciferin (purchased from PerkinElmer, Shanghai, China) was administered intravenously to the mice. Ten minutes later, anesthesia was induced via intraperitoneal injection of 300 mg/kg isoflurane. Three minutes following D-luciferin administration, *in vivo* bioluminescence imaging (BLI) was conducted using the IVIS® Lumina III *in vivo* imaging system (PerkinElmer, MA, USA) with an automatically adjusted exposure time. The acquired data were subsequently analyzed using Living Image software.

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantification

Tumor volume was measured using a vernier caliper to determine its length and width, and tumor size was calculated and evaluated based on the tumor volume formula: $1/2 \times a \times b^2$. The tumor growth inhibition rate was calculated using the formula: $(1 - TV_T / TV_C) \times 100\%$, where TV_C denotes the average tumor volume in the control group and TV_T denotes the average tumor volume in the treatment group. The percentage change in animal body weight was calculated using the formula: $\%BWC = [(BW_t - BW_0) / BW_0] \times 100\%$, where BW_t indicates the animal body weight at each measurement time point and BW_0 indicates the animal body weight at the baseline (grouping) time point.

Statistical analysis

All results are presented as the mean or mean $\pm$ standard deviation (SD). All statistical analyses were performed using GraphPad Prism 8.0.1, except that statistical power was calculated with IBM SPSS Statistics 27. Based on the experimental design, appropriate statistical tests including the unpaired Student's t-test and one-way analysis of variance (ANOVA) were employed. The statistical method employed is specified in the corresponding figure legend. A p-value of less than 0.05 was considered statistically significant with the following key: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.